NETFLIX CONTENT, RATINGS & ENGAGEMENT ANALYSIS

1. Project Overview

This project analyzes Netflix content and customer viewing behavior using data analytics and interactive Power BI visualizations. The analysis focuses on content availability, customer engagement, ratings, viewing behavior, subscription patterns and country-level performance.

2. Problem Statement

Analyze Netflix's content and customer viewing behavior to derive meaningful business insights. The goal is to clean and visualize the data to understand customer engagement, content popularity, and subscription trends.

3. Objectives

- Clean and prepare the Netflix dataset for analysis.
- Analyze content additions and content availability.
- Understand customer viewing behavior using watch-time metrics.
- Compare ratings across genres, months, countries, content types and watcher categories.
- Identify highly rated directors and popular content.
- Analyze subscription plans across countries.
- Identify countries with relatively lower engagement and ratings.
- Develop an interactive Power BI dashboard and derive business recommendations.

4. Tools and Technologies

The project was developed using Microsoft Excel and Microsoft Power BI, with DAX used within Power BI for calculations and analytical measures. These tools were used throughout the data preparation, analysis, visualization, and reporting stages.

4.1 Microsoft Excel

Microsoft Excel was used for the initial data preparation and cleaning process. The raw Netflix dataset was examined and prepared before being imported into Power BI.

Excel was used to:

- Inspect the structure and contents of the dataset.
- Identify and handle missing values.
- Check for duplicate records.
- Correct and standardize data formats.
- Verify numerical and categorical values.
- Perform basic data validation.
- Prepare the dataset for further analysis in Power BI.
- Perform preliminary calculations where required.

Excel provided a convenient environment for organizing and validating the raw data before dashboard development.

4.2 Microsoft Power BI

Microsoft Power BI was the primary tool used for data visualization, analysis, and interactive dashboard development.

Power BI was used to:

- Import the cleaned Excel dataset.
- Create relationships and organize data for analysis.
- Develop interactive dashboards.
- Create KPI cards and gauge charts for key performance indicators.
- Create bar charts for comparing ratings, genres, directors, countries, and watch time.
- Create line charts to analyze rating and content trends across months.
- Create scatter charts to examine relationships between engagement and ratings.
- Create pie/donut charts for subscription and customer-viewing distribution.

- Create maps to analyze country-level content and engagement.
- Add slicers for interactive filtering based on country, genre, age range, rating, subscription type, and content type.
- Apply Top N filters to identify highly rated directors and popular content.
- Develop separate dashboard pages for Content Analysis, Engagement Analysis, and Ratings Analysis.

4.3 DAX

Data Analysis Expressions (DAX) was used within Power BI to perform calculations and create calculated columns and measures.

DAX was used for:

- Calculating average ratings.
- Calculating total and average watch time.
- Counting titles and customers.
- Categorizing content as Newly Added or Older Content.
- Creating customer age categories.
- Creating watcher categories.
- Supporting comparisons between movies and TV shows.
- Analyzing country-level engagement and ratings.
- Creating measures that dynamically respond to dashboard filters and slicers.

For example:

Average Rating

Average Rating = AVERAGE('Netflix Dataset'[rating])

Total Watch Time

Total Watch Time = SUM('Netflix Dataset'[watch_time_minutes])

These DAX calculations helped convert the raw data into meaningful analytical metrics for the dashboard.

4.4 Overall Technology Workflow

The project followed the workflow:

Raw Netflix Dataset → Excel Data Cleaning & Preparation → Power BI Data Import → DAX Calculations → Data Visualization → Interactive Dashboard → Business Insights

5. Dataset Description

The project uses a Netflix customer/content dataset of approximately 20,000 records. Important fields used include show_id, title, type, genre, director, date_added, country, customer_name, age, age_range, subscription_type, watch_time_minutes and rating.

Variable	Purpose
show_id	Unique content identifier
title	Movie or TV show title
type	Movie or TV Show
genre	Content genre
director	Director of the content
date_added	Content addition date
country	Country used for regional analysis
watch_time_minutes	Customer engagement measure
rating	User rating
subscription_type	Basic, Standard or Premium
age_range	Customer age group

6. Data Cleaning and Preparation

- Checked missing and null values in important columns.
- Checked duplicate records and inconsistent categorical values.
- Converted date_added into a proper date format for month/year analysis and filtering.
- Ensured rating and watch_time_minutes were numeric.
- Standardized fields such as country, type, genre and subscription_type.

- Created age-range and watcher/content categories for analysis.
- Validated data types before building Power BI visuals.

7. Data Modeling and DAX

The cleaned dataset was loaded into Power BI. Aggregations such as Average Rating, Average Watch Time, Sum of Watch Time and Count of Title/Show ID were used in the visuals. Calculated categories were created to support comparison of newly added versus older content, age groups and watcher behavior.

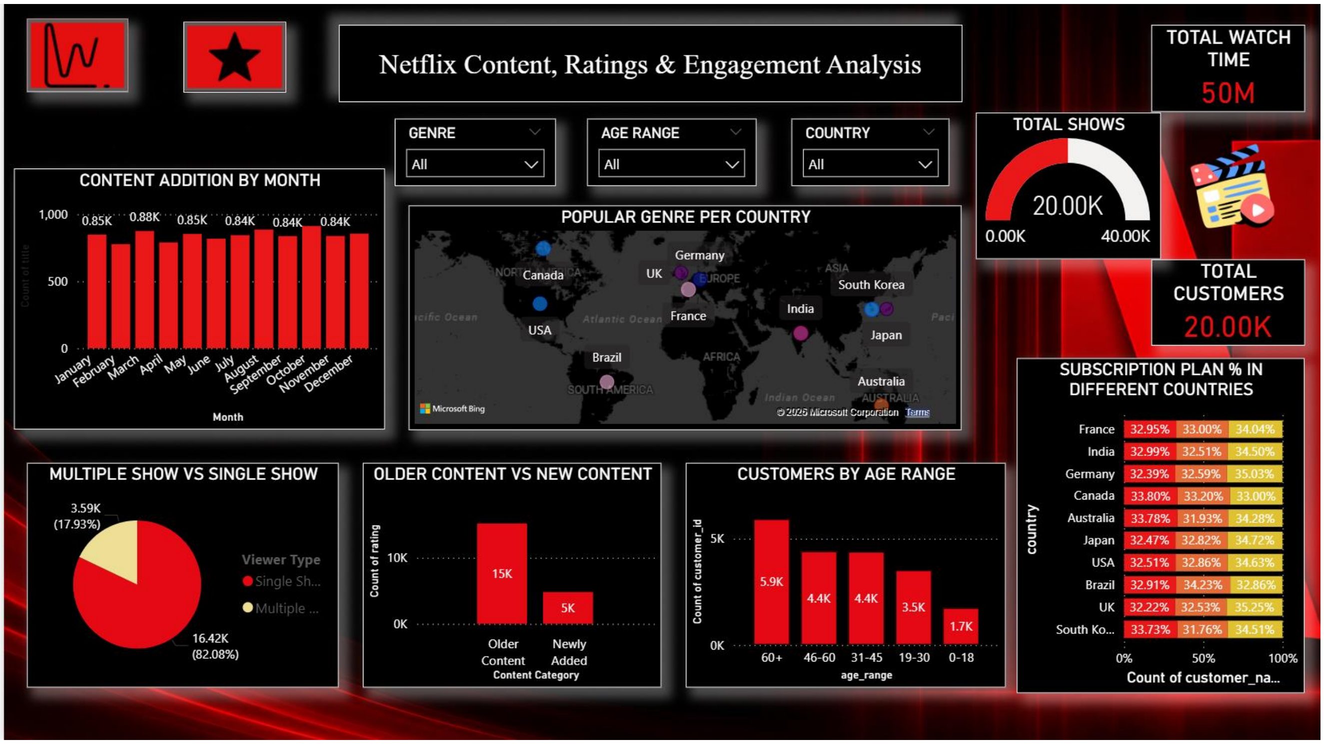
- Content Category – separates newly added content from older content.
- Age Range – groups customers into meaningful age bands.
- Watcher Category – separates heavier and lighter viewers based on the project logic.

8. Dashboard Design

A Netflix-inspired black, red and white theme was used. The report contains three analytical pages: Content Analysis, Engagement Analysis and Ratings. Slicers and navigation buttons make the dashboard interactive.

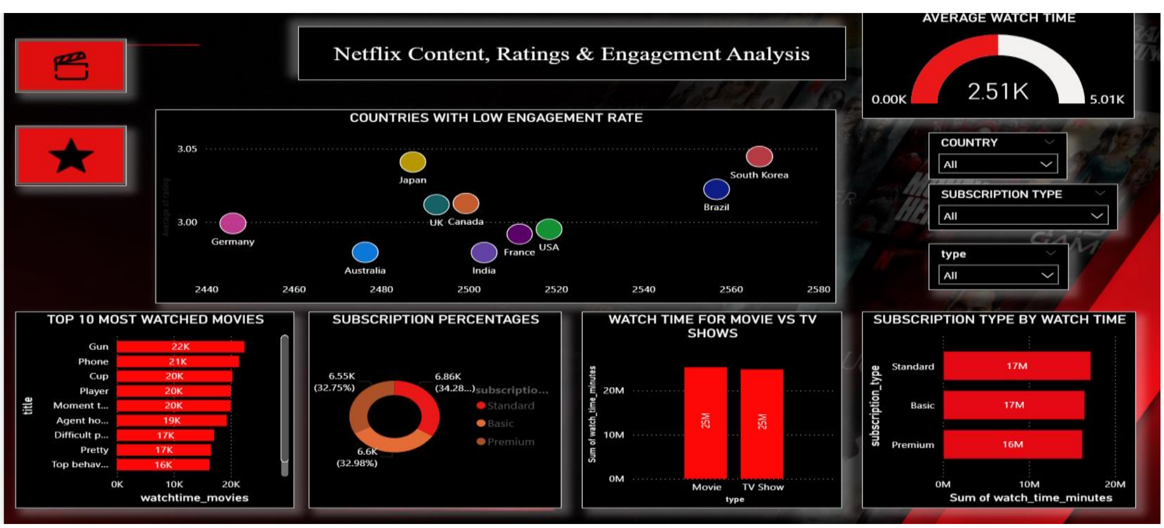
9. Content Analysis Dashboard

- **Content Addition by Month:** Shows the number of titles added in each month.
- **Popular Genre per Country:** Compares genre patterns across countries.
- **Total Shows:** Displays the overall content count.
- **Total Watch Time:** Displays total viewing time.
- **Total Customers:** Displays the number of customers represented.
- **Multiple Show vs Single Show:** Compares customers associated with multiple versus single shows.
- **Older Content vs New Content:** Compares older and newly added content.
- **Customers by Age Range:** Shows customer distribution across age groups.
- **Subscription Plan % in Different Countries:** Compares Basic, Standard and Premium shares by country.



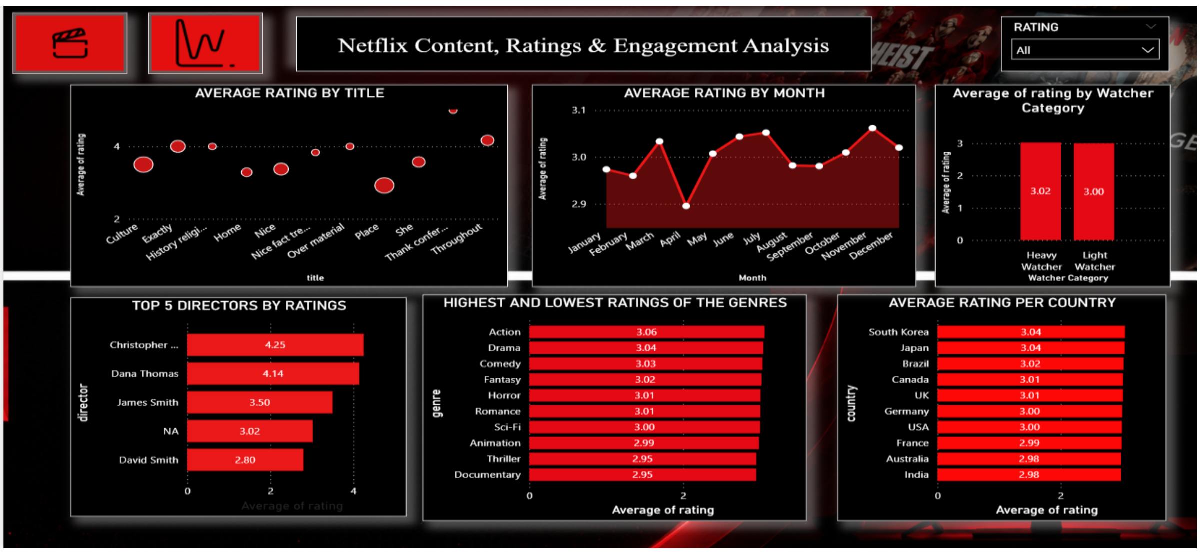
10. Engagement Analysis Dashboard

- **Countries with Low Engagement Rate:** Scatter/bubble analysis comparing average watch time and average rating by country.
- **Average Watch Time:** Gauge showing average viewing time.
- **Top 10 Most Watched Movies:** Ranks movies using total watch time.
- **Subscription Percentages:** Shows the distribution of subscription plans.
- **Watch Time for Movie vs TV Shows:** Compares total viewing time by content type.
- **Subscription Type by Watch Time:** Compares viewing time generated by subscription plans.



11. Ratings Analysis Dashboard

- **Average Rating by Title:** Compares average ratings across selected titles.
- **Average Rating by Month:** Identifies monthly differences in average ratings.
- **Average Rating by Watcher Category:** Compares ratings from heavy and light watchers.
- **Top 5 Directors by Ratings:** Ranks directors using average user rating and a Top N filter.
- **Highest and Lowest Ratings by Genre:** Identifies genres with higher and lower average ratings.
- **Average Rating per Country:** Compares average ratings across countries.



12. Key Findings

- Content additions remain substantial across the analyzed months, with month-to-month variation.
- The dashboard shows a larger share of single-show viewers than multiple-show viewers, suggesting an opportunity to encourage deeper catalog consumption.
- The displayed age analysis shows the 60+ group as the largest customer segment.
- Movie and TV Show watch time can be compared to determine which content format generates stronger engagement.
- The country scatter analysis shows that engagement and ratings vary by market.
- Action and Drama are among the higher-rated genres in the displayed ratings analysis.
- The Top 5 Directors visual identifies directors with comparatively high average user ratings.
- Monthly rating variation can help identify stronger periods for content releases.

13. Business Recommendations

- Increase localized content investment in lower-engagement markets.
- Promote high-rated genres and titles to customers based on their viewing preferences.
- Use personalized recommendations to convert single-show viewers into multiple-show viewers.
- Study successful directors and genres when planning future content.
- Use subscription-level watch-time patterns for targeted retention and upgrade campaigns.
- Consider release timing when planning major content launches.
- Monitor watch time and ratings together when evaluating regional and content performance.

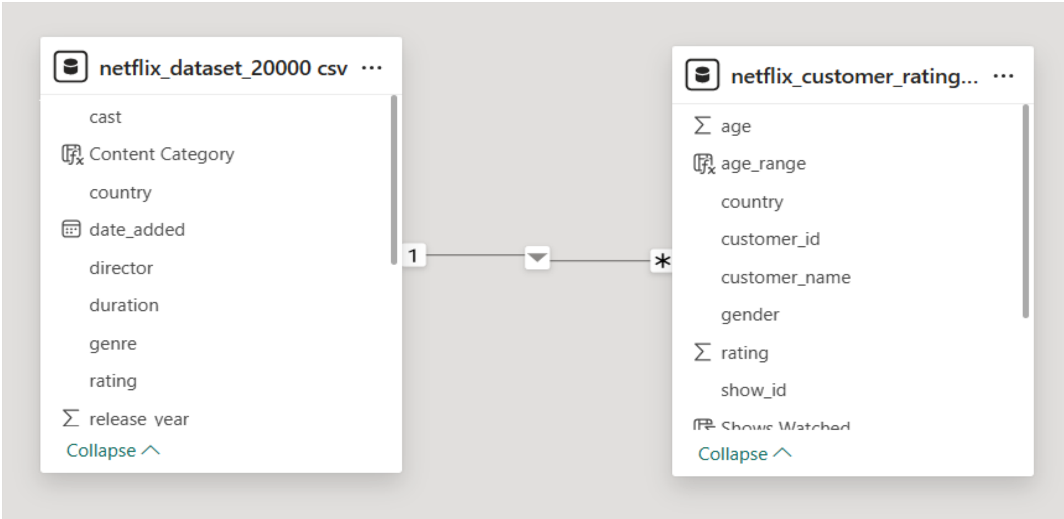
14. Conclusion

The Netflix Content, Ratings & Engagement Analysis project demonstrates how data cleaning, DAX calculations and interactive Power BI visualizations can transform raw customer and content data into useful business insights. The dashboard provides a consolidated view of content additions, customer engagement, ratings, subscriptions and regional performance. These insights can support content planning, customer targeting, regional strategies and engagement improvement.

15. Project Evaluation Metrics

- Data quality and consistency after cleaning.
- Correct use of measures, calculated columns and aggregations.
- Dashboard readability and visual consistency.
- Interactivity through slicers and navigation.
- Coverage of content, engagement, ratings and subscription analysis.
- Business relevance of findings and recommendations.

ustomer_id	show_id	customer_name	age	gender	country	subscription_type	watch_time_minutes	rating	age_range	Shows Watched	Viewer Type	Watcher Cate
5008	s17930	Mrs. Angela Ochoa	36	Other	Brazil	Standard	3189	5	31-45	1	Single Show	Heavy Watcl
5042	s15386	Karen Brady	72	Other	UK	Standard	3943	5	60+	1	Single Show	Heavy Watcl
5057	s15066	Bryan Clark	61	Other	Japan	Standard	92	5	60+	1	Single Show	Light Watche
5080	s7103	Stephen Owens	36	Other	Germany	Standard	567	5	31-45	1	Single Show	Light Watche
5122	s7472	Cory Acosta	78	Other	France	Standard	2110	5	60+	1	Single Show	Light Watche
5132	s6622	Cassie Torres	26	Other	USA	Standard	613	5	19-30	1	Single Show	Light Watche
5187	s4120	Ralph Young	24	Other	India	Standard	3046	5	19-30	1	Single Show	Heavy Watcl
5244	s18606	Shawn Hill	33	Other	Canada	Standard	113	5	31-45	2	Multiple Shows	Light Watche
5258	s18032	Dr. Brandon Gonzalez DDS	31	Other	Japan	Standard	291	5	31-45	1	Single Show	Light Watche
5299	s9089	William Carson	31	Other	South Korea	Standard	785	5	31-45	1	Single Show	Light Watche
5331	s15669	Lisa Perry	28	Other	South Korea	Standard	2210	5	19-30	1	Single Show	Light Watche
5335	s2468	Angela Taylor	48	Other	Brazil	Standard	2891	5	46-60	3	Multiple Shows	Light Watche
5344	s18007	Heather Landry	48	Other	UK	Standard	4104	5	46-60	1	Single Show	Heavy Watcl
5371	s8636	April Dunn	22	Other	Brazil	Standard	3135	5	19-30	1	Single Show	Heavy Watcl
5395	s7921	Joshua Jackson	37	Other	South Korea	Standard	3749	5	31-45	2	Multiple Shows	Heavy Watcl
5414	s12957	Kevin Alvarez	75	Other	Brazil	Standard	2040	5	60+	1	Single Show	Light Watche
5468	s7978	Keith Anderson	66	Other	USA	Standard	4737	5	60+	1	Single Show	Heavy Watcl
5513	s9446	Lawrence Peterson	37	Other	France	Standard	4090	5	31-45	1	Single Show	Heavy Watcl
5537	s3640	Rebekah Bautista	27	Other	Japan	Standard	84	5	19-30	1	Single Show	Light Watche
5543	s3365	Julie Yu	18	Other	Canada	Standard	4360	5	0-18	1	Single Show	Heavy Watcl
5589	s15698	Christopher Scott	80	Other	UK	Standard	843	5	60+	2	Multiple Shows	Light Watche


netflix_dataset_20000 csv
cast
Content Category
country
date_added
director
duration
genre
rating
release year
Collapse
1
*
netflix_customer_rating...
age
age_range
country
customer_id
customer_name
gender
rating
show_id
Collapse